

Haozhe (Hank) Si

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Research Interests

COMPUTER VISION: Multi-Modal Foundation Models, Remote Sensing, Generative Models

MACHINE LEARNING: Parameter Efficient Transfer Learning, OOD Generalization

Education

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

URBANA, IL, UNITED STATES

Ph.D., *Electrical and Computer Engineering* (GPA: 4.0)

08/2022-05/2027

- Advisor: Prof. Han Zhao

Bachelor of Science, *Computer Engineering* (GPA: 3.97)

08/2018-12/2021

Publications

- [1] **H. Si**, B. Zhao, D. Wang, M. Chen, Y. Gao, Z. Wang, X. Li, “Fully Self-Supervised Depth Estimation from Defocus Clue”. In *Proceedings of the Computer Vision and Pattern Recognition (CVPR 2023)*
- [2] **H. Si**, X. Zhang, Z. Zhang, “Furry IP-Adapter: Preserving the Text-to-Image Capability of Stable Diffusion in IP-Adapter”. In *Amazon Computer Vision Conference Workshop on Multimodal Generation (ACVC 2024)*
- [3] H. Wang, **H. Si**, B. Li, H. Zhao, “Provable domain generalization via invariant-feature subspace recovery”. In *Proceedings of the International Conference on Machine Learning (ICML 2022)*
- [4] H. Wang, **H. Si**, H. Shao, H. Zhao, “Enhancing Compositional Generalization via Compositional Feature Alignment”. In *Transactions on Machine Learning Research (TMLR)*
- [5] Y. Chen, **H. Si**, G. Zhang, H. Zhao, “Moment Alignment: Unifying Gradient and Hessian Matching for Domain Generalization”. In *Proceedings of Conference on Uncertainty in Artificial Intelligence (UAI 2025)*
- [6] **H. Si**, Y. Wan, M. Do, V. Deepak, H. Zhao, H. F. Hamann, “Towards a Foundation Model for Multi-modal and Multi-spectral Geospatial Data”. In *TerraBytes Workshop at the International Conference on Machine Learning (TerraBytes -ICML 2025)*
- [7] **H. Si**, Y. Wan, Y. Wang, M. Do, H. Zhao, “LESSViT: Robust Hyperspectral Representation Learning under Spectral Configuration Shift”. In *Proceedings of Neural Information Processing Systems (NeurIPS 2026)*

Pre-prints

(* indicates equal contribution.)

- [8] **H. Si**, Y. Wan, Y. Wang, M. Do, H. Zhao, “ChronoEarth-492K: A Large Scale and Long Horizon Spatiotemporal Hyperspectral Dataset and Benchmark”. Under Review for *ISPRS Journal of Photogrammetry and Remote Sensing*
- [9] H. Wang*, G. Balasubramaniam*, **H. Si**, B. Li, H. Zhao, “Invariant-Feature Subspace Recovery: A New Class of Provable Domain Generalization Algorithms”. Under Review for *Journal of Machine Learning Research (JMLR)*

Research Experiences

IBM-IL DISCOVERY ACCELERATOR INSTITUTE (IIDAI)

Aug. 2023-Aug. 2026

Research Assistant. With Prof. Han Zhao and Dr. Hendrick Hamann.

- Conducted research on designing flexible and efficient foundation models for multi-modal and hyperspectral geospatial data.
- Conducting research on developing foundation models for spatiotemporal and multi-source geospatial data.
- Developed large-scale geospatial datasets and data pipelines involving multi-source data acquisition, spatial and temporal indexing, geospatial alignment, metadata management, quality control, and scalable pre-processing.
- Developing a large-scale event-centric InSAR dataset, including event-based data discovery, geospatial sampling, temporal organization, preprocessing, and quality assessment of interferometric observations.

CUSTOMER TRUST, AMAZON.COM INC.

May. 2025-Aug. 2025

Applied Scientist Intern. With Prof. Rui Song, Prof. Hengrui Cai and Prof. Sheng Wang

- Conducted research on conditioned text-embedding synthesis via Autoregressive Conditional Variational Autoencoder.

MERCH ON DEMAND, AMAZON.COM INC.

May. 2024-Aug. 2024

Applied Scientist Intern. With Dr. Zhixiong Zhang

- Conducted research on enhancing the text-controllability and identity-fidelity for personalized image generative models.

SHANGHAI AI LAB

Mar. 2022-Sep. 2023

Research Intern. With Prof. Bin Zhao.

- Conducted research on fully self-supervised Depth from Defocus (DfD).

Service

REVIEWER

- Advances in Neural Information Processing Systems (NeurIPS) 2023,2026
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2023, 2024, 2026
- AAAI Conference on Artificial Intelligence (AAAI) 2024
- Transactions on Machine Learning Research (TMLR) 2024
- Computer Vision and Pattern Recognition (CVPR) 2026
- European Conference on Computer Vision (ECCV) 2026

Honors

- James Scholar, the Grainger College of Engineering, University of Illinois
- The O. Thomas and Martha S. Purl Scholarship
- Highest Honors at Graduation, the Grainger College of Engineering, University of Illinois

Teaching Experience

- ECE 385: Digital Systems Laboratory Spring 2023, Fall 2023
- ECE 544: Pattern Recognition Fall 2026